

Business Requirements Document (BRD)

Project: Aarvii CCTV AMC Management System **Phase:** D0 — Project Foundation Documentation **Source of truth:** requirements-freeze-v1.md (Approved & Frozen, V1)

Every requirement below traces to a freeze-document section (cited as §n). This BRD adds structure and IDs only — no new requirements, features, or rule changes.

1. Business objectives

ID	Objective	Freeze ref
BO-1	Capture and convert website-generated leads into AMC customers	§2, §10
BO-2	Centralize customers, sites, contacts, and asset summaries	§5–§7
BO-3	Manage the full AMC contract lifecycle with preserved renewal history	§8, §9
BO-4	Automate and verify preventive maintenance service delivery	§11–§13
BO-5	Resolve complaints through a tracked ticket lifecycle	§14
BO-6	Bill customers per AMC contract term with self-service invoice access	§16, §19
BO-7	Provide customer and engineer self-service on web and mobile	§2, §18
BO-8	Reuse the existing base platform — no duplicate implementation	§20

2. Functional requirements

2.1 Public Website (§2, §10)

ID	Requirement
FR-WEB-01	The system shall serve a public website at www.aarvii.in with pages: Home, About Us, Services, AMC Services, Contact Us, Gallery, Testimonials, Login.
FR-WEB-02	The website shall reuse existing public website content wherever possible.
FR-WEB-03	The website shall provide a Get Quote enhancement allowing visitors to request quotations.
FR-WEB-04	The website shall provide an AMC Inquiry enhancement allowing visitors to submit AMC inquiries.
FR-WEB-05	Website inquiries (contact, AMC inquiry, quote requests) shall automatically create leads (§10).

2.2 Lead Management (\$10)

ID	Requirement
FR-LEAD-01	The system shall manage leads through the statuses: New, Contacted, Qualified, Quotation Sent, Negotiation, Won, Lost, Converted.
FR-LEAD-02	Converting a lead shall create a Customer, a Site, and an Initial AMC Contract.
FR-LEAD-03	A notification shall be sent on Lead Created and Lead Converted (§17).

2.3 Customer Management (§3, §5)

ID	Requirement
FR-CUST-01	Admin shall manage customer records.
FR-CUST-02	A customer shall own one or more sites (§6).
FR-CUST-03	Customers shall be able to update their own profile and reset their password (OTP-based, §17).

2.4 Site Management (§5, §6)

ID	Requirement
FR-SITE-01	A site shall belong to exactly one customer.
FR-SITE-02	A site shall support a maximum of three contact persons.
FR-SITE-03	A site shall hold at most one active AMC contract at a time.
FR-SITE-04	A site shall aggregate: contact persons, asset summary, AMC contract, scheduled visits, tickets, and invoices (§5).

2.5 Asset Management (§7)

ID	Requirement
FR-ASSET-01	Assets shall be tracked as summary counts per site : Camera, DVR, NVR, Hard Disk, Switch, Router, Monitor.
FR-ASSET-02	The system shall not track individual cameras/devices.
FR-ASSET-03	Asset summaries may optionally record Brand, Model, and Remarks.

2.6 AMC Plans (§9)

ID	Requirement
FR-PLAN-01	Admin shall manage AMC plans (e.g. Silver, Gold, Platinum).
FR-PLAN-02	A plan shall store: Price, Visit Frequency, Included Services, SLA.
FR-PLAN-03	Plans shall be versioned ; modifying a plan shall never change historical contracts.

2.7 AMC Contracts (§8)

ID	Requirement
FR-AMC-01	AMC contracts shall follow a Master + Terms model: a permanent contract record plus renewal-term history records.
FR-AMC-02	Customers shall see the current active term; Admin shall see the complete renewal history.
FR-AMC-03	Customers shall be able to request AMC renewal (§3).
FR-AMC-04	An AMC Expiry Reminder notification shall be sent (§17).
FR-AMC-05	The system shall generate an AMC Contract PDF (§19).

2.8 Service Scheduling (§11)

ID	Requirement
FR-SCHED-01	Visits shall be automatically generated from AMC plan visit frequency.
FR-SCHED-02	Visit schedule statuses shall be: Planned, Assigned, In Progress, Completed, Missed, Cancelled.
FR-SCHED-03	Admin shall be able to reschedule visits.
FR-SCHED-04	Engineer assignment shall be mandatory before a visit proceeds.
FR-SCHED-05	Visit Scheduled and Visit Completed notifications shall be sent (§17).

2.9 Visit Management (§12, §13)

ID	Requirement
FR-VISIT-01	Visit completion shall require: engineer selfie, GPS coordinates, minimum one visit photo, customer signature, and visit remarks.
FR-VISIT-02	GPS capture shall store latitude, longitude, and timestamp.
FR-VISIT-03	Visits shall support Before, During, and After photos.
FR-VISIT-04	Engineers shall submit visit reports for admin review ; admin approval is required before a customer can view a report.
FR-VISIT-05	The system shall generate a Visit Report PDF (§19).
FR-VISIT-06	Engineers shall be able to upload photos, videos, and reports (§15).

2.10 Ticket Management (§14)

ID	Requirement
FR-TKT-01	Tickets shall follow the statuses: Open, Assigned, In Progress, Resolved, Closed, Reopened.
FR-TKT-02	Tickets shall carry a priority: Low, Medium, High, Critical.
FR-TKT-03	Customers, Admins, and Engineers shall be able to create tickets (engineers during visits, §3).

ID	Requirement
FR-TKT-04	Customers shall be able to reopen closed tickets.
FR-TKT-05	Ticket Created, Ticket Assigned, and Ticket Closed notifications shall be sent (§17).

2.11 Engineer Management (§15)

ID	Requirement
FR-ENG-01	Admin shall manage engineers.
FR-ENG-02	Engineers shall view their assigned work (visits and tickets).
FR-ENG-03	Engineers shall not manage customers, AMC plans, or contracts.

2.12 Invoice Management (§16)

ID	Requirement
FR-INV-01	Invoices shall follow the statuses: Draft, Generated, Sent, Paid, Cancelled.
FR-INV-02	Every invoice shall be linked to an AMC Contract Term.
FR-INV-03	Customers shall be able to download invoices as PDF (§19).
FR-INV-04	An Invoice Generated notification shall be sent (§17).
FR-INV-05	No accounting features shall be included in V1.

2.13 Notifications (§17)

ID	Requirement
FR-NOTIF-01	The system shall send notifications via Email and SMS .
FR-NOTIF-02	Notification events: Lead Created, Lead Converted, Ticket Created, Ticket Assigned, Ticket Closed, Visit Scheduled, Visit Completed, AMC Expiry Reminder, Invoice Generated, Password Reset OTP, Login OTP.

2.14 Reporting (§2)

ID	Requirement
FR-RPT-01	The Admin Portal shall provide a Reporting module covering the managed business areas.

2.15 Customer Portal (§2)

ID	Requirement
FR-CP-01	The Customer Portal (web + mobile) shall provide: Dashboard, AMC Details, Service History, Upcoming Visits, Tickets, Invoices, Profile Management, Password Reset.

2.16 Engineer Portal (§2)

ID	Requirement
FR-EP-01	The Engineer Portal (web + mobile) shall provide: Assigned Visits, Assigned Tickets, Visit Reporting, Photo Upload, GPS Capture, Selfie Capture, Customer Signature, Ticket Creation.

2.17 Mobile applications (§18)

ID	Requirement
FR-MOB-01	Mobile apps shall be built with Flutter .
FR-MOB-02	Customer App features: Dashboard, AMC, Tickets, Invoices, Notifications, Profile.
FR-MOB-03	Engineer App features: Visits, Tickets, Photo Upload, GPS Capture, Signature Capture, Offline Support.

2.18 PDF documents (§19)

ID	Requirement
FR-PDF-01	The system shall generate: AMC Contract PDF, Visit Report PDF, Invoice PDF.

3. Non-functional requirements

Derived from the freeze document and the mandated platform reuse (§20); platform-provided qualities reference the [platform discovery report](#).

ID	Requirement	Basis
NFR-01	Security/Auth — All portals authenticate via the platform Auth module (JWT/OAuth2); role-based access enforces the four-actor permission model.	§3, §20
NFR-02	Authorization — Engineer restrictions (§15) and customer data ownership ("own contracts/invoices") are enforced server-side via platform RBAC/ABAC.	§3, §15
NFR-03	Tenancy & data isolation — Customer-facing data is scoped so customers access only their own sites, contracts, visits, tickets, and invoices.	§3
NFR-04	Auditability — Business actions are captured by the platform Audit capability (visit approval, ticket transitions, contract changes available for audit).	§20
NFR-05	File handling — Photos, videos, selfies, signatures, and PDFs are stored via the platform Files capability (tenant-scoped, provider-pluggable).	§12, §19, §20
NFR-06	Notifications delivery — Email via the platform Notifications capability; SMS via an SMS provider integration.	§17, §20
NFR-07	Offline capability — The Engineer App must operate offline and synchronize when connectivity returns.	§18
NFR-08	Mobile platform — Both apps use the existing Flutter Mobile Foundation (secure token storage, push/notification plumbing, release engineering).	§18, §20
NFR-09	Historical integrity — Plan versioning guarantees historical contracts never change when plans are modified.	§9

ID	Requirement	Basis
NFR-10	Evidence integrity — GPS records latitude, longitude, and timestamp; mandatory completion evidence cannot be bypassed.	\$12
NFR-11	Observability — Logging, tracing, correlation IDs, and health checks are inherited from the platform Host.	\$20
NFR-12	Documentation governance — All CCTV modules follow the platform documentation governance (7-file module docs, ADRs, docs validation CI).	Platform policy

4. Business constraints

ID	Constraint	Source
BC-01	Scope is frozen; changes only via approved change request.	\$22
BC-02	The Core Platform is frozen at v1.0.0 — CCTV functionality ships as business modules; no Core modifications.	\$20, platform freeze policy
BC-03	No duplicate implementation of Authentication, Roles, Permissions, Files, Notifications, Audit, Theme Engine, Mobile Foundation, API Keys, Webhooks.	\$20
BC-04	Existing public website content must be reused wherever possible.	\$2
BC-05	One active AMC contract per site; max 3 contact persons per site.	\$6
BC-06	Assets tracked as summary counts only.	\$7
BC-07	No accounting features in invoicing.	\$16
BC-08	Out-of-scope list (\$21) must not be implemented in V1.	\$21
BC-09	Web frontend is React 19 + Theme Engine; mobile is Flutter — per the base platform stack.	\$20, platform

5. Assumptions

ID	Assumption
AS-01	Aarvii Technologies operates as a single tenant of the platform; multi-tenancy isolation still applies between customer accounts at the application level.
AS-02	Existing website content (copy, images, testimonials) is available for reuse (\$2).
AS-03	An SMS gateway provider will be selected and contracted for SMS notifications (\$17); email uses the platform provider model.

ID	Assumption
AS-04	Engineers have Android/iOS smartphones with camera and GPS for the Engineer App (§12, §18).
AS-05	Customer signature is captured on the engineer's device screen during the visit (§12).
AS-06	Visit frequency semantics (e.g. visits per term) are defined per AMC plan (§9) and drive auto-generation (§11).
AS-07	Invoice amounts derive from the AMC plan/term pricing; payment recording is a manual status update (no payment gateway, §21).

6. Dependencies

ID	Dependency	Needed by
DEP-01	Ashraak Platform V1 (frozen) — Auth, Roles/Permissions, Files, Notifications (email), Audit, Theme Engine, Mobile Foundation, API Keys, Webhooks	All modules (§20)
DEP-02	SMS gateway provider	Notifications (§17)
DEP-03	Email provider (SMTP) configuration	Notifications (§17)
DEP-04	PDF generation capability (server-side)	AMC Contract / Visit Report / Invoice PDFs (§19)
DEP-05	Existing www.aarvii.in content	Public Website (§2)
DEP-06	Device capabilities: camera, GPS, touch signature	Engineer App (§12, §18)
DEP-07	App store accounts + platform release pipelines (fastlane CI)	Mobile distribution (§18)

7. Traceability

- All FRs cite their freeze-document section inline.
- Business rules are enumerated separately in [business-rules.md](#).
- Workflows: [workflow-overview.md](#) · Design: [high-level-design.md](#)
- Screens: [screen-inventory.md](#) · Navigation: [navigation-map.md](#)